

Internship Report

Topic

IMS Cloud Security

Organized by

KGSL Group

Venue

Jyothi Engineering College, Thrissur

Duration

1 Week

Focus Area

AWS Cloud Services - Fundamentals and Security

1. Introduction

This report summarizes the one-week internship program on IMS Cloud Security, conducted by KGSL Group at Jyothi Engineering College, Thrissur. The internship provided a practical and conceptual introduction to Amazon Web Services (AWS), with a special emphasis on cloud infrastructure and security components.

2. Objectives of the Internship

- To introduce students to cloud computing and AWS fundamentals.
- To provide hands-on training in AWS services like EC2, S3, RDS, and IAM.
- To explore key security aspects and architectural configurations in cloud systems.
- To promote understanding of real-world cloud infrastructure management.

3. Internship Structure

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The internship spanned six working days, each day covering essential aspects of cloud computing and AWS services.

Day 1: Introduction to Cloud Computing

- Basics of cloud computing
- Cloud service models: IaaS, PaaS, SaaS
- Cloud deployment models: Public, Private, Hybrid
- Overview of major cloud providers

Day 2: AWS Console and Core Services

- Navigating the AWS Management Console
- Introduction to EC2 (Elastic Compute Cloud)
- Basics of launching and managing instances
- Overview of AWS global infrastructure

Day 3: Networking and VPC Setup

- Introduction to Amazon VPC (Virtual Private Cloud)
- Configuring subnets, route tables, and security groups
- Creating and launching EC2 instances within a VPC
- Understanding availability zones and regions

Day 4: Amazon S3 and Target Groups

- Introduction to Amazon S3 (Simple Storage Service)
- Creating and configuring S3 buckets
- S3 access control and permissions
- Introduction to load balancing and Target Groups in AWS
- Setting up target groups for EC2 instances

Day 5: Amazon RDS and VPC Peering

- Introduction to Amazon RDS (Relational Database Service)
- Creating and managing RDS instances

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- Connecting RDS with applications
- Understanding VPC Peering and enabling private communication between VPCs

Day 6: IAM and Cloud Security Fundamentals

- Overview of AWS Identity and Access Management (IAM)
- Creating IAM users, roles, groups, and policies
- Implementing least privilege access
- AWS shared responsibility model
- Best practices for securing cloud infrastructure

4. Key Takeaways

- Acquired foundational knowledge of AWS services and cloud concepts.
- Gained practical skills in deploying and configuring cloud infrastructure.
- Understood how to manage storage using S3, and configure load balancing using target groups.
- Learned how to set up and connect to RDS instances, and establish secure communication via VPC peering.
- Gained a thorough understanding of IAM and cloud security best practices.

5. Conclusion

The internship offered a comprehensive introduction to cloud computing with a strong emphasis on AWS tools and secure cloud architecture. Each session combined practical implementation with theoretical insight, preparing participants for further exploration and certification in cloud technology. The experience has enhanced my confidence in working with cloud platforms and inspired me to pursue deeper specialization in cloud security and AWS solutions.

6. Acknowledgements

I would like to sincerely thank KGSL Group for conducting this insightful and well-organized internship. I also extend my gratitude to Jyothi Engineering College, Thrissur, for hosting the program and providing the necessary infrastructure. Special thanks to the instructors and support staff for their dedication and guidance.

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throughout the internship.